# Tools — MasterFetch

MasterFetch provides rich web content extraction, multi-engine search, and same-domain crawling with a content cache. Outputs carry envelope signals (`content_ok`, `page_type`, `next_action`, `source_type`, `is_official`, `content_age_days`, `is_stale`).

| Tool | Description |
|---|---|
| `mf_fetch` | Rich URL/PDF extraction with envelope signals. |
| `mf_search` | Keyless multi-engine web search. |
| `mf_crawl` | Best-first same-domain crawl. |
| `mf_cache_clear` | Clear the content cache. |
| `mf_screenshot` | Capture a page as a screenshot. **Registered only when a drivable headless browser engine is present** (spec `openhands` FR-029); absent from the default build. |
| `mf_version` | Return integration version info. |

**Use cases:** extracting article content, searching multiple engines in parallel, crawling a domain.

`mf_search` runs DuckDuckGo, Brave, OpenAlex, and Wikipedia in parallel (plus optional LangSearch / Tavily / Perplexity / Exa / Serper engines when their keys are configured). `exclude_engines` removes named backends before any request is dispatched; naming every configured engine returns an explicit "all engines excluded" result. An optional `engine` restricts the search to a single backend.

Set `respect_robots: true` on `mf_fetch` / `mf_crawl` when unsure whether a site allows crawling; do not hammer the same domain with repeated calls.

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## mf_fetch

Fetch any URL or PDF with automatic content extraction.

**Arguments**

| Argument | Type | Required | Description | Typical value |
|---|---|---|---|---|
| `url` | string | one of `url`/`urls` | Single URL to fetch | `"https://example.com/article"` |
| `urls` | array | one of `url`/`urls` | Array of URLs for parallel bulk fetch (up to 8 concurrent) | `["https://a.com","https://b.com"]` |
| `format` | enum | no | `markdown` (default), `html`, `text`, `raw` | `"markdown"` |
| `css_selector` | string | no | Narrow extraction scope | `"article"` |
| `focus` | string | no | Query string for BM25 post-extraction filtering (no re-fetch) | `"memory safety"` |
| `max_content_chars` | integer | no | Maximum content characters (min 500) | `40000` |

| Argument | Type | Required | Description | Typical value |
|---|---|---|---|---|
| `offset` | integer | no | Character offset to resume pagination | `40000` |
| `include_links` | boolean | no | Classify outgoing links into citations/navigation/external | `false` |
| `respect_robots` | boolean | no | Check robots.txt before fetching | `false` |
| `cache_ttl` | integer | no | Cache TTL in seconds (`0` bypasses cache) | `3600` |

**Example:**

```
mf_fetch url="https://example.com" format="markdown"
mf_fetch urls=["https://a.example.com","https://b.example.com"] max_content_chars=20000
```

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## mf_search

Keyless multi-engine web search; engines run in parallel and results are merged, deduplicated, and ranked with `relevance_score`, `fetch_relevance`, and `engines_consensus`.

**Arguments**

| Argument | Type | Required | Description | Typical value |
|---|---|---|---|---|
| `query` | string | yes | Search query | `"zero-copy parsing in rust"` |
| `engine` | enum | no | Restrict to one backend: `openalex`, `wikipedia`, `langsearch`, `tavily`, `perplexity`, `exa`, `serper` | — |
| `exclude_engines` | array | no | Engine names to drop before dispatch | `["openalex"]` |
| `site` | string | no | Restrict results to a domain | `"rust-lang.org"` |
| `exclude_sites` | array | no | Domains to exclude | `["pinterest.com"]` |
| `freshness` | enum | no | `day`, `week`, `month`, `year` | `"week"` |

| Argument | Type | Required | Description | Typical value |
|---|---|---|---|---|
| `published_after` | string | no | Oldest admissible publication date as `YYYY-MM-DD`. An exact window the coarse `freshness` buckets cannot express; engines that accept an absolute lower bound apply it server-side (Exa `startPublishedDate`, Tavily `start_date`, Serper `cdr:1,cd_min:`), and it takes precedence over `freshness` for them. A value that is not `YYYY-MM-DD` is dropped with a warning rather than sent as a literal query parameter | `"2026-09-13"` |
| `max_results` | integer | no | Cap after merge/dedup (1–500) | `6` |
| `per_engine_results` | integer | no | Cap per engine before merge (1–200) | `75` |
| `page` | integer | no | Result page (0–10) | `0` |

**Example:**

```
mf_search query="rust async patterns" max_results=10
mf_search query="rust async patterns" exclude_engines=["openalex"]
mf_search query="site:doc.rust-lang.org ownership rules" site="doc.rust-lang.org"
mf_search query="frontier model pricing" published_after="2026-09-13"
```

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## mf_crawl

Best-first, same-domain crawl starting from a URL. Each page is returned as markdown with `content_ok` and `page_type` envelope signals.

**Arguments**

| Argument | Type | Required | Description | Typical value |
|---|---|---|---|---|
| `url` | string | yes | Start URL | `"https://docs.rs/serde"` |

| Argument | Type | Required | Description | Typical value |
|---|---|---|---|---|
| `max_pages` | integer | no | Maximum pages to fetch | `10` |
| `max_depth` | integer | no | Maximum crawl depth from start URL | `2` |
| `max_total_chars` | integer | no | Total character budget across pages | `200000` |
| `deadline_ms` | integer | no | Time budget in milliseconds | `120000` |
| `focus` | string | no | Query string for scoring/filtering pages | `"tutorial"` |
| `sitemap` | boolean or `"auto"` | no | Use sitemap (`true`), use if available (`"auto"`), or pure BFS (`false`) | `false` |
| `discover_only` | boolean | no | Return discovered URL map without fetching content | `false` |
| `crawl_urls` | array | no | Second-phase selective crawl: fetch only this subset of URLs | — |
| `respect_robots` | boolean | no | Check robots.txt before fetching | `false` |

**Example:**

```
mf_crawl url="https://doc.rustup.rs" max_pages=15 max_depth=2 focus="getting started"
```

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## mf_cache_clear

Clear the masterfetch content cache.

**Arguments**

| Argument | Type | Required | Description | Typical value |
|---|---|---|---|---|
| `all` | boolean | no | `true` wipes all entries; `false`/omitted purges only expired entries | `false` |

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## mf_screenshot

Capture a page as a screenshot image. This tool is registered **only when a drivable headless browser engine is present** (spec `openhands` FR-029): the default build compiles no driver, so the tool is absent from the advertised tool set and the capability is never offered when it cannot work. When a bundled driver is enabled (the opt-in `headless-browser` Cargo feature) and an engine binary resolves on `PATH`, the tool registers and returns a screenshot (or an honest error recommending `mf_fetch` for text-based extraction).

**Arguments**

| Argument | Type | Required | Description | Typical value |
|---|---|---|---|---|
| `url` | string | yes | URL to capture | `"https://example.com"` |
| `width` / `height` | integer | no | Viewport size in pixels | `1280` / `800` |
| `full_page` | boolean | no | Capture the full scrollable page | `false` |

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## mf_version

Return the masterfetch integration version, the ragent version, and a brief description of the tool set. Makes no network calls and always succeeds.

**Arguments:** none.